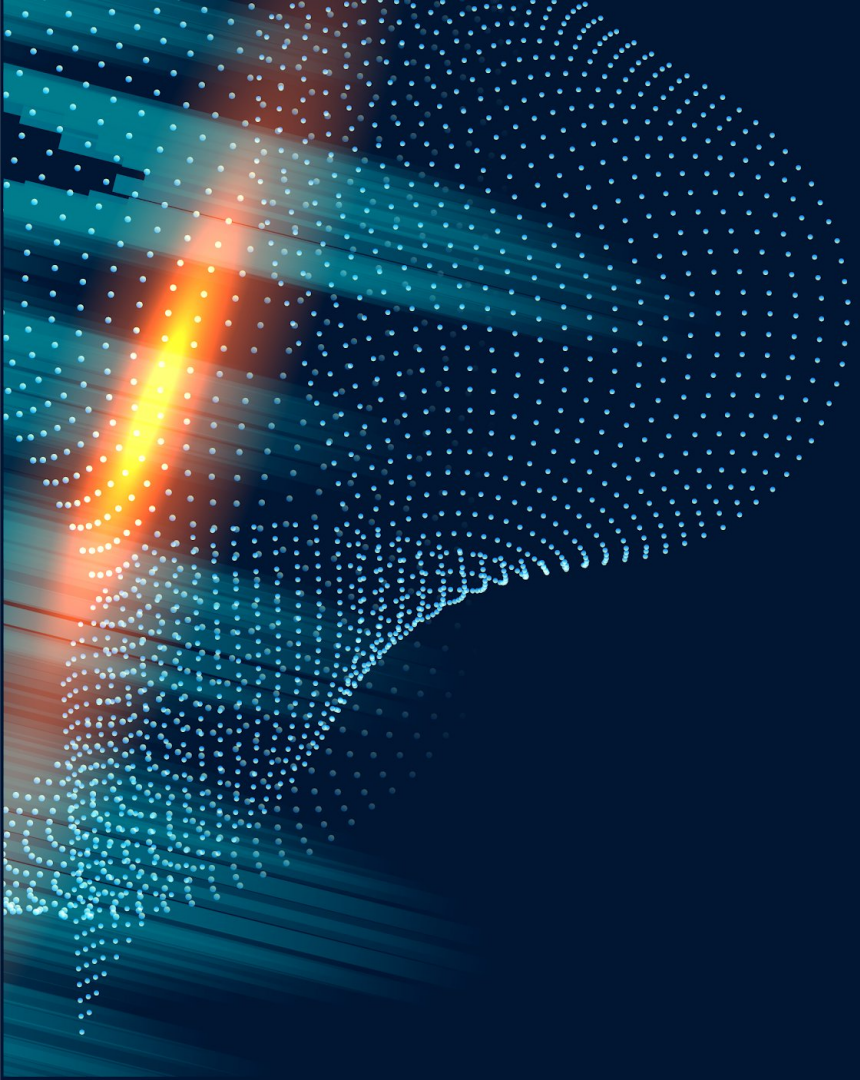
University of Tahri Mohamed-Béchar
Faculty of Exact Sciences
Dept. of Mathematics and Computer Sciences

Critical Thinking and Creativity Skills

Analytical Thinking & Argument Evaluation

Amina Bourouis, PhD

4INGCS, S7



Why This Matters for a Security Engineer?

- **Incident Response**: Is this a real attack or a false positive? Analyze the evidence.
- **Risk Assessment**: How do you prioritize patches? You must evaluate the arguments for and against each action.
- **Policy & Proposals:** "We should adopt a Zero-Trust model." You need to build a logical, persuasive argument to get budget approval.
- **Forensics:** Piecing together events from log files requires structured analytical reasoning.

What is Analytical Thinking?

Definition: The process of breaking down complex information into smaller, manageable components to understand its structure and meaning.

The Analytical Thinking Process :

1. **Identify** the problem or question.
2. **Deconstruct** it into its core parts.
3. **Examine** the relationships between the parts.
4. **Interpret** the information and draw inferences.
5. **Synthesize** a conclusion or solution.

The Building Blocks of an Argument

Definition: An argument is not a fight! It's a set of reasons (premises) offered in support of a claim (conclusion).

Key Components:

Conclusion: The main point the speaker/writer is trying to prove.

Indicator Words: Therefore, so, thus, hence, as a result, consequently, this proves that.

Premises : The evidence, reasons, or facts offered to support the conclusion.

Indicator Words: Because, since, for, given that, as, due to the fact that, the evidence is.

Evaluating Arguments - The Core Questions

Once you've identified an argument, you must evaluate its strength.

The Three Pillars of a Strong Argument:

- **Are the Premises Logically Connected to the Conclusion? (Logic)**
- **Are the Premises Actually True? (Truth)**
- **Is the Argument Complete? (Are there hidden assumptions or missing information?)**

Logical Structure (Validity & Soundness)

- **Logical Validity**: IF the premises are true, does the conclusion **necessarily** follow?
  - *Valid Example*: "All phishing emails have links. This email has a link. Therefore, this is a phishing email." (This is valid in form but...)
- **Soundness**: An argument is sound only if it is valid and all its premises are true.
  - Is the first premise true? No. Not **all** phishing emails have links; some have attachments. Therefore, the argument is valid but unsound.

Truth of Premises (Evidence & Credibility)

- **Questions to Ask :**
  - **What is the source?**: Is it a peer-reviewed paper, a vendor's marketing blog, an anonymous forum post?
  - Is the evidence current? A report on attack vectors from 2010 is likely obsolete.
  - Is it biased? A company selling a new AI-powered security tool might overstate the risks that their product solves.
  - Can it be verified? Can you find independent sources that confirm the same finding?

Hidden Assumptions & Missing Information

- **Hidden Assumption**: An unstated premise that is necessary for the conclusion to hold.
- Example Argument: "We don't need to encrypt the internal database because it's behind the corporate firewall."

Hidden Assumptions & Missing Information

Stated Premise: The database is behind the corporate firewall.
Conclusion: We don't need to encrypt it.
Hidden Assumptions:

1. The corporate firewall is 100% impenetrable.
2. All internal users are trustworthy (no insider threat).
3. Other attack vectors from inside the network don't exist.

Common Logical Fallacies in Security Discussions

1. Appeal to Fear (FUD)
2. False Dilemma (Either/Or)
3. Appeal to Authority
4. Anecdotal Fallacy
5. Slippery Slope

Key Takeaways

- Analytical Thinking is your methodology for deconstructing complex security problems.
- Arguments are built from Premises that lead to a Conclusion.
- To Evaluate an argument, ask:
  - Is it logical? (Do the premises support the conclusion?)
  - Is it true? (Are the premises based on solid evidence?)
  - Is it complete? (What is being assumed or left out?)
  - Your role as a security engineer is to be the analytical skeptic, ensuring decisions are based on sound reasoning, not just fear, tradition, or convenience.

THANKS!

Do you have any questions?

bourouisamina@gmail.com

Slide credits : slidesgo, freepik, flaticon